Mohammed Mahroos

Software Engineer Intern · Full-Stack Developer

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SUMMARY

Information Technology student at the University of Moratuwa – Institute of Technology (ITUM) with practical experience building full-stack web applications, REST APIs, and relational database systems using Java, Spring Boot, React, FastAPI, and PostgreSQL. Hands-on with real-time updates (WebSocket), JWT-secured APIs, Docker-based deployment, and AI service integration. Seeking a Software Engineer Intern to contribute clean, reliable code and grow within a collaborative engineering team.

PROFESSIONAL EXPERIENCE

Entry-Level Software Developer Present

Independent Developer

- Acquired practical software development experience through self-directed study and hands-on projects.
- Built skills in frontend and backend technologies to support full-stack application development.
- Strengthening problem-solving, debugging, and analytical abilities through practical coding tasks.
- Used version control tools, standard development workflows, and database management systems.
- continuously expanding my technical knowledge in the fields of software engineering, networking, cloud computing, and DevOps..

EDUCATION

National Diploma in Technology (NDT) 2025 – Present

University of Moratuwa – Institute of Technology (ITUM)

GCE Advanced Level – Mathematic, Physics, Chemistry 2021 – 2023

P/President Science Collage- - Puttalam

TECHNICAL SKILLS

Languages	Java · JavaScript · Python · PHP
Frontend	React.js · Next.js · HTML5 · CSS · Tailwind CSS · Vite
Backend	Spring Boot · Node.js · WebSocket · FastAPI
Database	PostgreSQL · MySQL · SQL
Mobile App	Flutter · Angular
Tools	IntelliJ · VS Code · Figma · Docker · Docker Compose · Postman
Version Control	Git · GitHub

LANGUAGES

English Tamil Sinhala

PROJECTS

Smart Municipal Council Management System | Link

- Built a full-stack civic platform for complaint tracking, service requests, sanitation oversight, admin dashboards, and PayHere online payments.
- Developed a Spring Boot (Java 17) REST API with Spring Security, JWT, role-based access control, file uploads, and PostgreSQL via Spring Data JPA.
- Implemented a React 19 frontend with Vite, Tailwind CSS, Chart.js dashboards, and Leaflet / Google Maps integration.
- Integrated a Python AI microservice using the Gemini API for complaint cleaning, classification, duplicate detection, priority scoring, and a citizen chatbot.
- Containerized the frontend, backend, AI service, and PostgreSQL using Docker Compose with automated SQL schma and seed script initialization.

EFL Season 2 – Football League Management System |Link

- Developed a full-stack football league management system with a FastAPI backend, a Next.js live-score website, and a Next.js administration console.
- Built REST APIs for tournaments, teams, players, fixtures, lineups, standings, statistics, and in-game match events.
- Implemented WebSockets to stream live scores and real-time match events to the public website.
- Designed and integrated a PostgreSQL database to store tournament data, team records, player profiles, match events, and statistics.

Hospital Appointment Management System |Link

- Developed a Spring Boot backend with MySQL to manage patient, doctor, and appointment data.
- Implemented CRUD operations for patient and doctor records.
- Built REST APIs for appointment scheduling. Added validation logic to prevent double booking.
- Used object-oriented programming principles and a layered architecture to keep the codebase modular and maintainable.

ERP Finance Management Module |Link

- Designed PostgreSQL database schemas for users, budgets, expenses, invoices, and payroll.
- Wrote SQL queries and configured data relationships to support core financial operations.
- Built the payroll calculation logic to automate net salary computation.

CERTIFICATIONS

Python Essentials 2025
Cisco Networking Academy · OpenEDG Python Institute

Python for Beginners 2025
CODL, University of Moratuwa · Verification Code: TTirvE5X2F

Introduction to Python 2025
SoloLearn · Certificate ID: CC-UVJ6RIGO

ACHIEVEMENTS

University Colours – Football 2025
University of Moratuwa · Awarded for loyalty and performance

Second Place – Volleyball (Under 15) 2016
Divisional Sports Meet, Puttalam Educational Zone · P/Al-Aqsa N.S.

REFERENCES

Uthpala Athukorala
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